TP-AGB Stars can Undergo Stable Roche Lobe Overflow Mass Transfer

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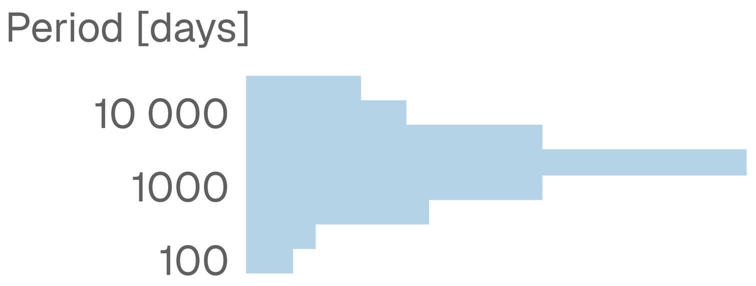
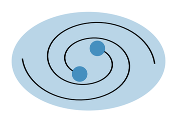
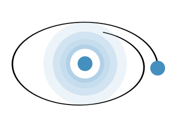
Mass transfer from thermally pulsing (TP) asymptotic giant branch (AGB) stars can produce barium stars, which are often observed with periods between 100 and 20 000 days.[1,2]



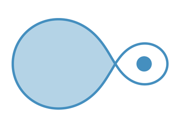
The period distribution of barium stars is difficult to explain:



Common Envelope Events result in orbits that are too small.



Wind mass transfer cannot explain the small orbits.



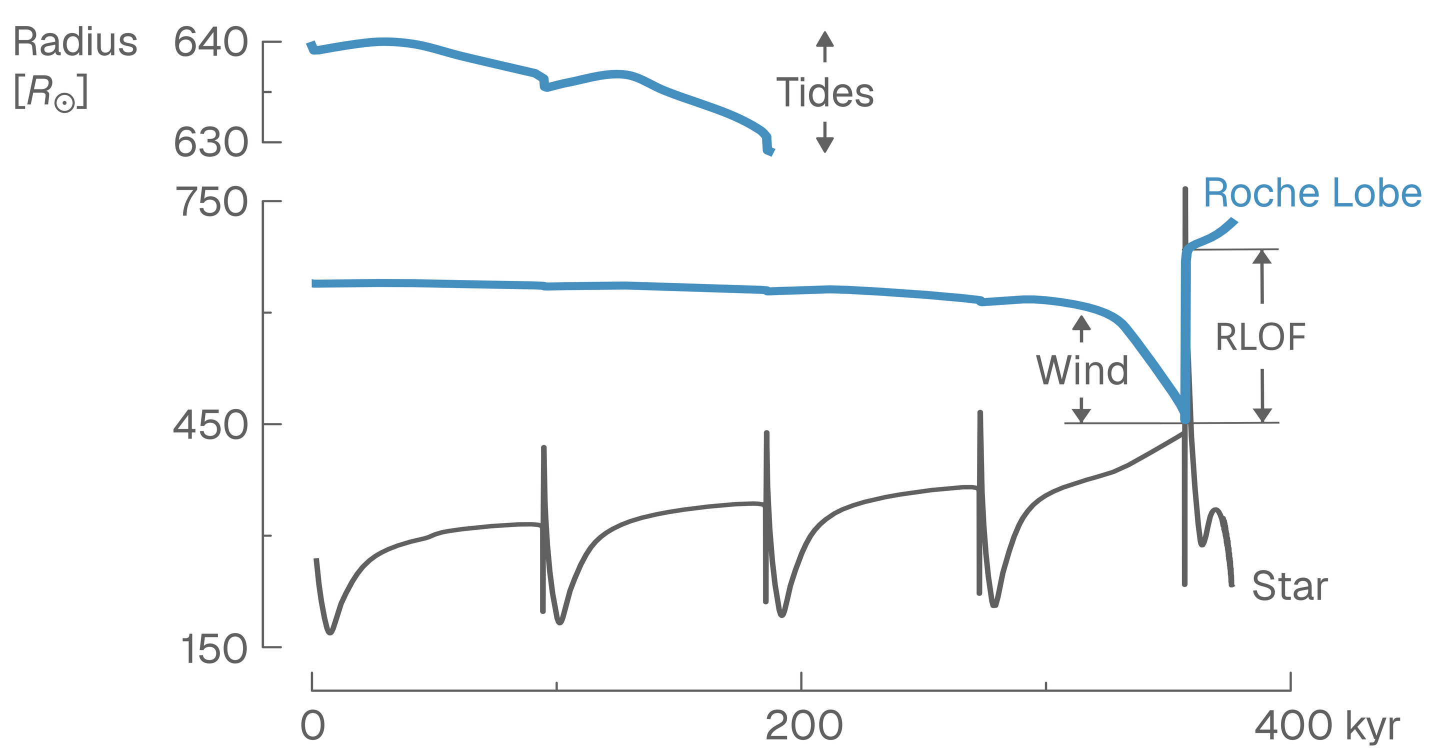
Roche lobe overflow (RLOF) mass transfer is often assumed unstable for giants, *but* recent models[3] show many RGB an E-AGB stars transfer mass stably.

In our project, we test if TP-AGB stars can experience stable RLOF.

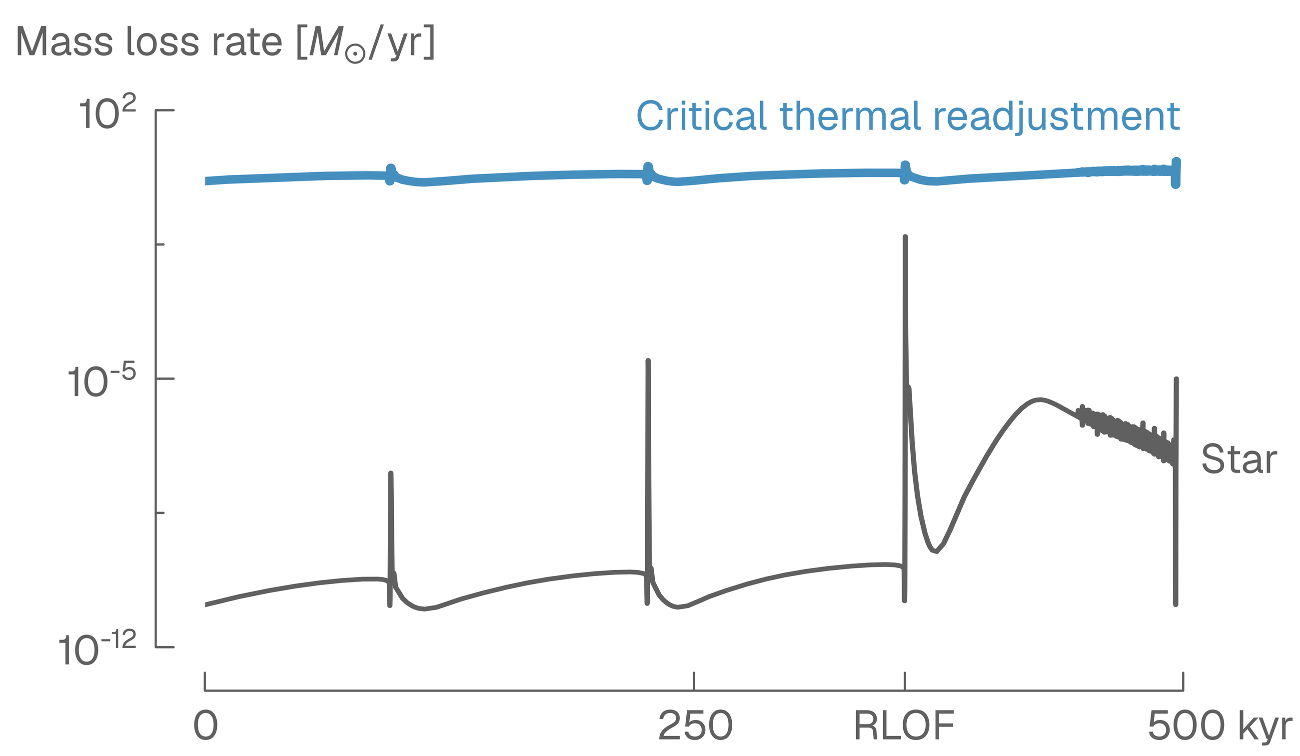
We simulate conservative RLOF from TP-AGB stars using *MESA*, and include dredge-up[4], wind mass transfer[5], and tides[6].

We evaluate the stability of RLOF and analyse the orbital effects.

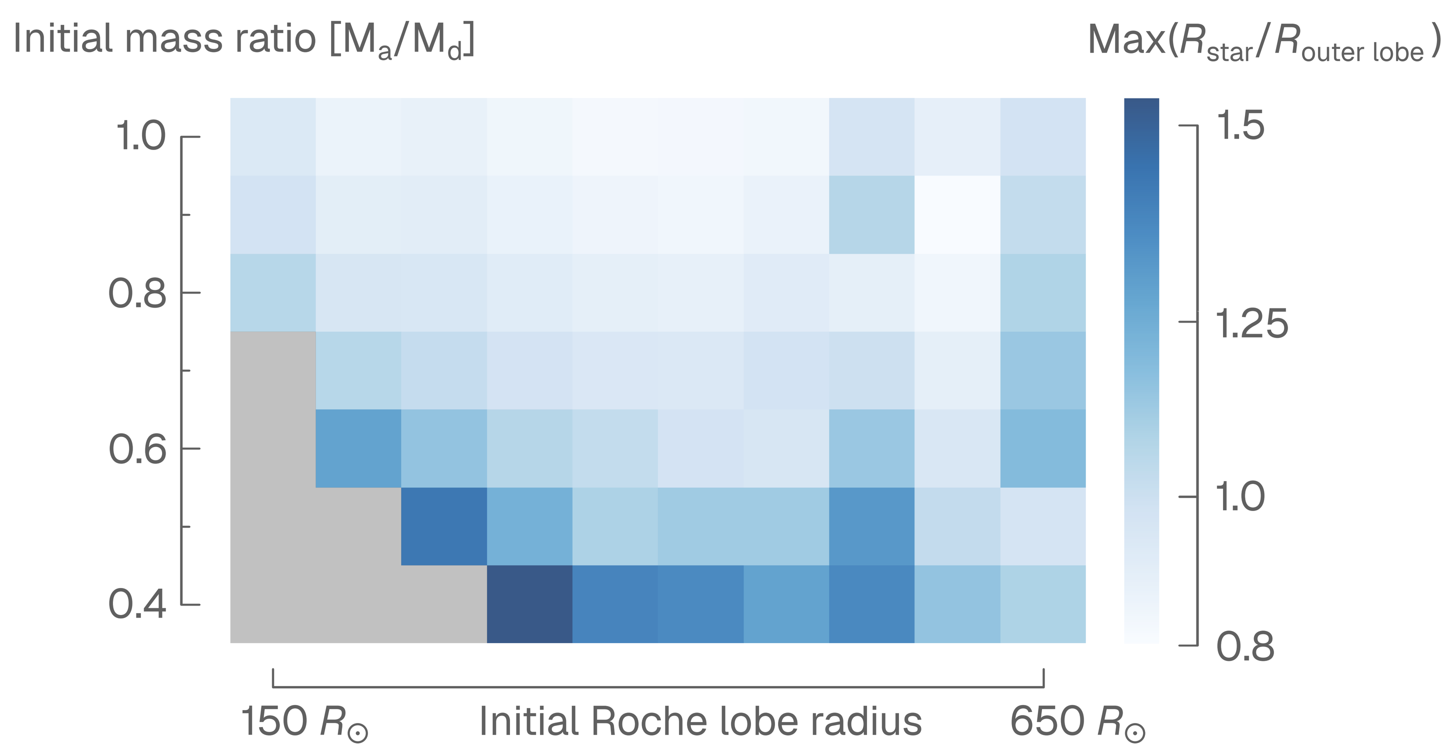
Stellar evolution and winds drive systems into RLOF.



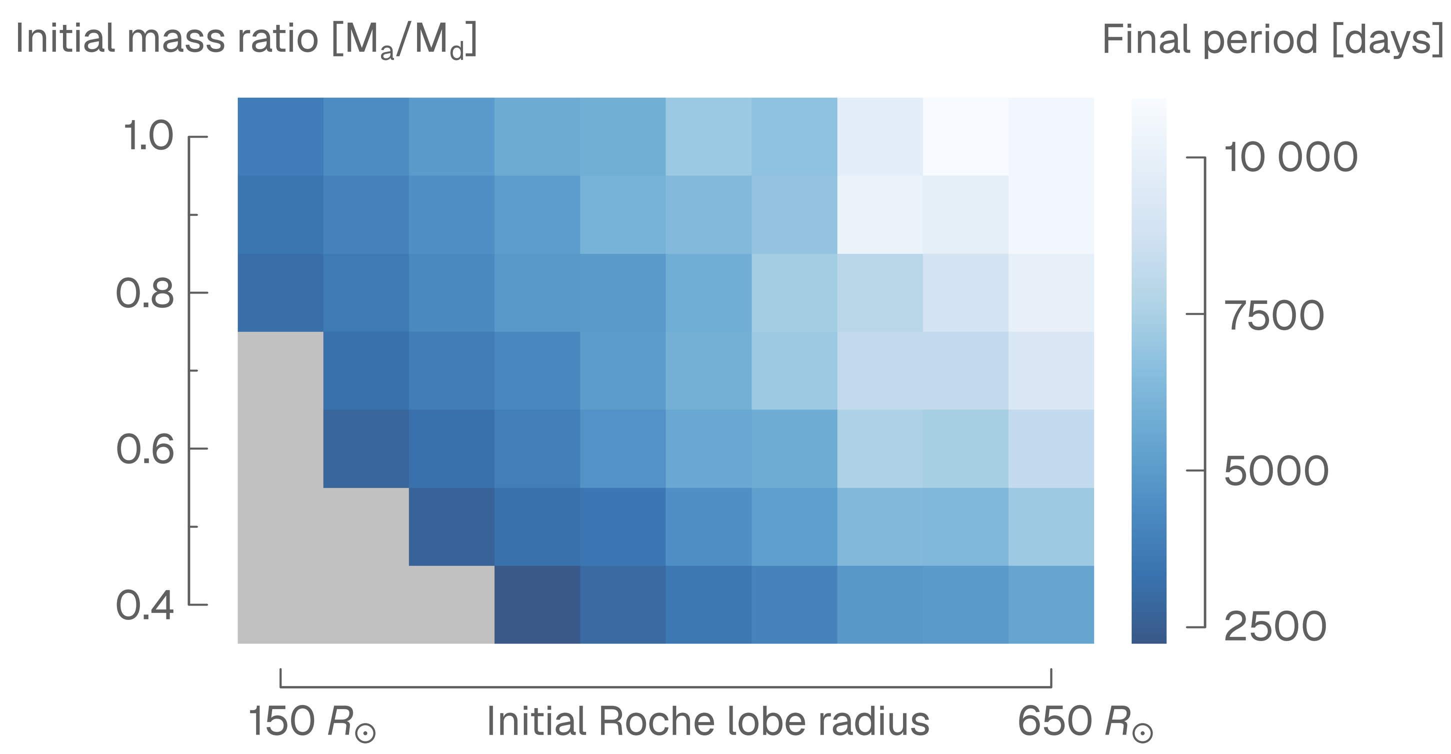
The surface layers of all our TP-AGB donor stars can thermally readjust to RLOF before being stripped away.



The TP-AGB stars reach radii close to, or in some cases, larger than their outer Roche lobe radius during RLOF.



Fully conservative RLOF mass transfer results in periods larger than those of the observed barium star population.



In the future, we want to:

- Explore non-conservative mass transfer and mass loss from outer Lagrange points.
- Investigate mass transfer in systems with non-zero eccentricity.
- Thoroughly compare our results from stable mass transfer to properties of barium stars.

[1]: Jorissen, A., Boffin, H. M. J., Karinkuzhi, D., et al. (2019) A&A626, A127.
[2]: Escorza, A. & De Rosa, R. J. (2023) A&A671, A97.
[3]: Temmink, K. D., Pols, O. R., Justham, S., Istrate, A. G., & Toonen, S. (2023) A&A669, A45.
[4]: Rees, N. R., Izzard, R. G., & Karakas, A. I. (2024) MNRAS527, 9643.
[5]: Saladino, M. I., Pols, O. R., van der Helm, E., Pelupessy, I., & PortegiesZwart, S. (2018) A&A618, A50
[6]: Preece, H. P., Hamers, A. S., Neunteufel, P. G., Schaefer, A. L., & Tout, C. A.(2022) ApJ933, 25.